AEPSC Aransas Pass to Rincon 69-kV Line Rebuild Project – ERCOT Independent Review Scope

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RPG Meeting
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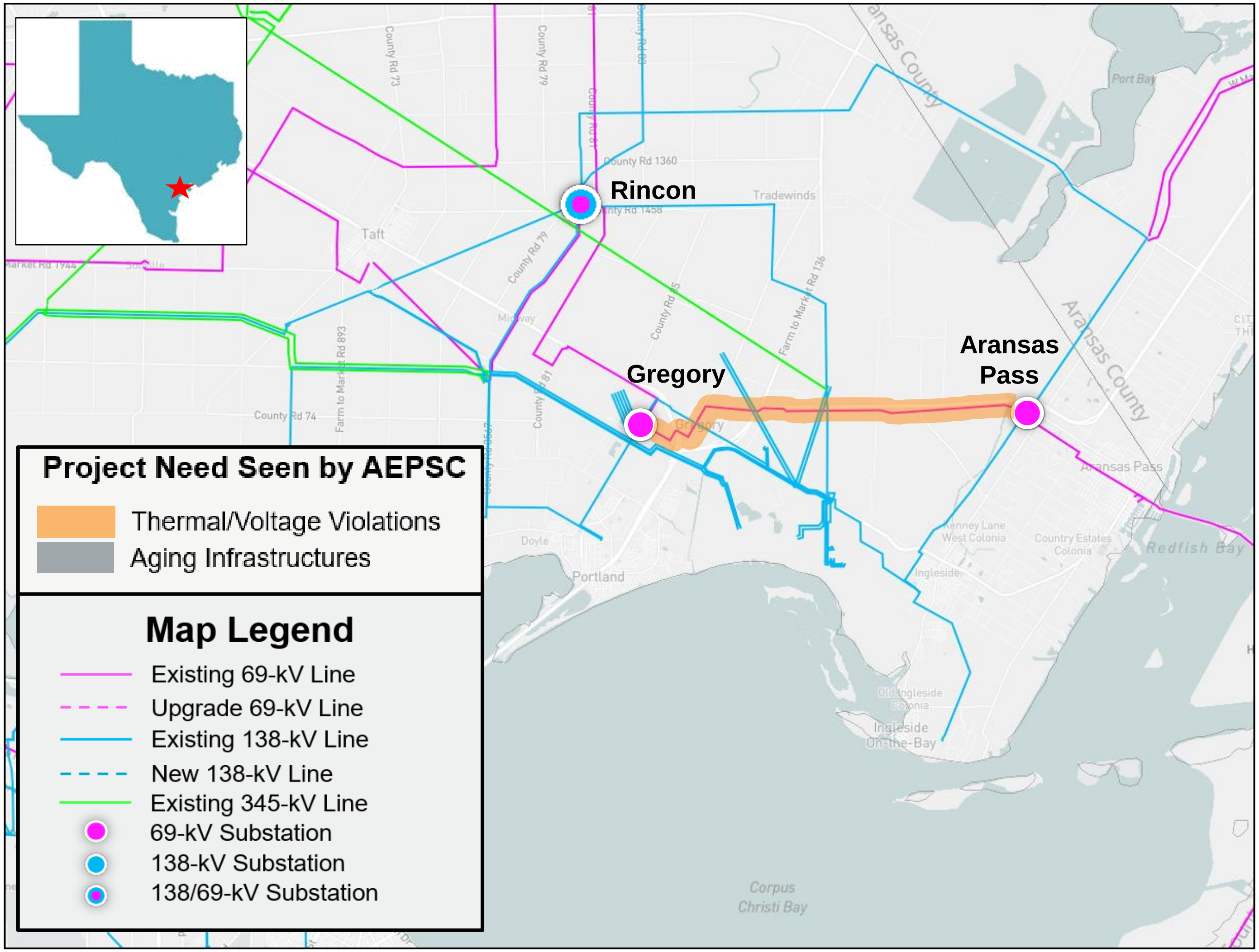
Introduction

- Aransas Pass to Rincon 69-kV Line Rebuild Project for Regional Planning Group (RPG) review in November 2024
  - This is a Tier 2 project with an estimated cost of $33.0 million and will require a Certificate of Convenience and Necessity (CCN)
  - Estimated in-service date is June 2026
  - This project is needed to address post-contingency thermal overloads in the San Patricio County
- This project is currently under ERCOT Independent Review (EIR)

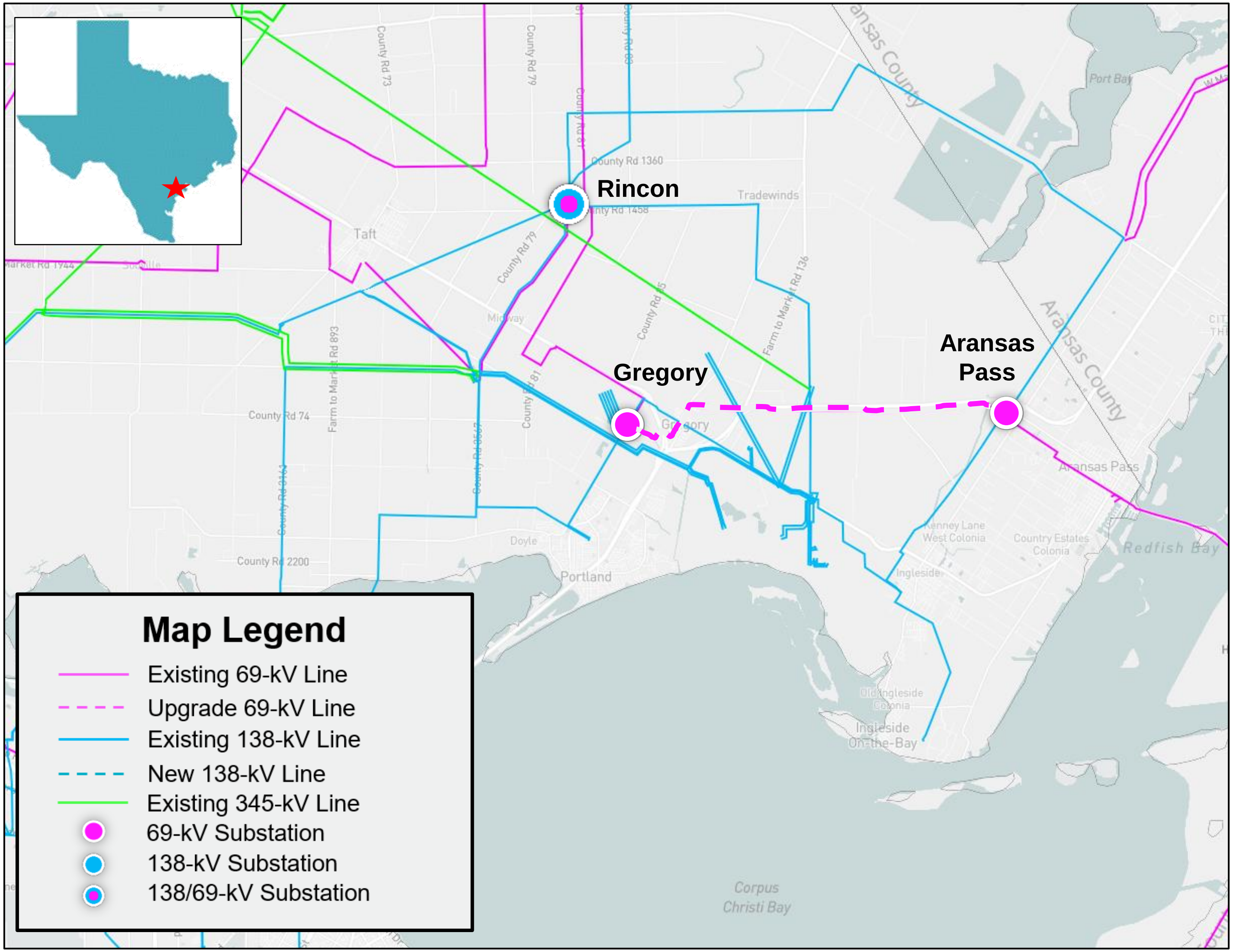
Study Area Map with Project Need Seen by AEPSC



Proposed Project by AEPSC

- Rebuild the existing Aransas Pass – Gregory 69-kV transmission line to 138-kV capable, but operational at 69-kV, normal & emergency ratings of at least 239 MVA, approximately 8.5-mile
- Rebuild the existing Gregory – Rincon 69-kV transmission line to 138-kV capable, but operational at 69-kV, normal & emergency ratings of at least 239 MVA, approximately 0.03-mile
- Upgrade the existing Gregory substation to at least 2,000 A capable station. Replace bus-tie switch at Gregory with bus-tie breaker
- Upgrade the existing Gregory 69-kV transmission line terminal at Aransas Pass to at least 2,000 A capability
- Upgrade Gregory 69-kV line terminal at Rincon to at least 2,000 A capability

Map with Proposed Project by AEPSC



Study Assumptions – Base Case

- Study Area
  - San Patricio County in South Weather Zone is considered for this study
  - Aransas, Jim Wells, and Nueces Counties which are electrically close to project location
- Steady-State Base Case
  - Final 2023 Regional Transmission Planning (RTP) 2026 summer peak load case for South and South Central (SSC) Weather Zones, published on Market Information System (MIS) in December 2023, will be updated to construct the study base case
    - Case: 2023RTP_2026_SUM_SSC_12222023
    - Link: https://mis.ercot.com/secure/data-products/grid/regional-planning

Study Assumptions – Transmission

- Based on the October 2024 Transmission Project and Information Tracking (TPIT) posted on the ERCOT website projects within the study area with in-service dates before June 2026 within the study area will be added to the study base case if not already modeled in the case
  - TPIT Link: https://www.ercot.com/gridinfo/planning

TPIT	Project Name	Tier	Project ISD	County(s)
71940	LCRATSC_Saxet_Substation_Addition	Tier 4	5/01/24	Nueces

- Transmission projects identified in the 2023 RTP in the study area that have not been approved by RPG will be removed from the study base case

RTP Project ID	Project Name	County(s)
2023-S1	LGE (160205) to Dupont Switch - Ingleside (8422) Ckt 1 138-kV Upgrade	AEPSC

Study Assumptions – Generation

- There is no new generation that met Planning Guide Section 6.9(1) condition with Commercial Operation Date (COD) before the June 2026 in the study area at the time of the study not already modeled in the RTP case based on November 2024 Generator Interconnection Status (GIS) report published in MIS in December 2024
  - Link: https://www.ercot.com/gridinfo/resource
- All generation will be dispatched consistent with the 2024 RTP methodology
- All recent retired/indefinitely mothballed units will be reviewed and opened (turned off), if not already reflected in the 2023 RTP Final cases

Study Assumptions – Load & Reserve

- Loads in study area
  - Load level in the study area was maintained consistent with the Final RTP cases
  - Additional new approved loads in the study area will be added to the study base case
- Reserve
  - The reserve was maintained consistent with the 2023 RTP

Contingencies & Criteria

- Contingencies
  - NERC TPL-001-5.1 and ERCOT Planning Criteria
  - (http://www.ercot.com/mktrules/guides/planning/current)
    - P0 (System Intact)
    - P1, P2-1, P7 (N-1 conditions)
    - P2-2, P2-3, P4, and P5 (345-kV only)
    - P3 (G-1+N-1: G-1 of Midway Wind, Nueces Bay Repower Stg 7, and Papalote Creek Wind II)
    - P6-2 (X-1+N-1: X-1 of Lon Hill and Whitepoint 345/138-kV transformers)
- Criteria
  - Monitor all 69-kV and above busses, transmission lines, and transformers in the study region (excluding generator step-up transformers)
  - Thermal
    - Use Rate A for normal conditions
    - Use Rate B for emergency conditions
  - Voltage
    - Voltages exceeding their pre-contingency and post-contingency limits
    - Voltage deviations exceeding 8% on non-radial load buses

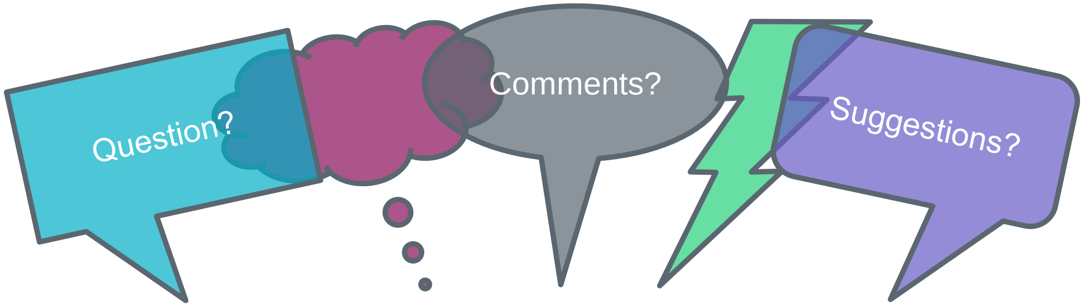
Study Procedure

- Need Analysis
  - The reliability analysis will be performed to identify the need to serve the projected area load using the study base case
- Project Evaluation
  - Project alternatives will be tested to satisfy the NERC and ERCOT reliability requirements
  - ERCOT may also perform the following studies
    - Planned maintenance outage
    - Long-Term Load-Serving Capability Assessment
  - TSP(s) will provide Cost and Feasibility Assessment
- Congestion Analysis
  - Congestion analysis may be performed based on the recommended transmission upgrades to ensure that the identified transmission upgrades do not result in new congestion within the study area

Deliverables

- Tentative Timelines
  - Status updates at the future RPG meetings
  - Final recommendation – Q1 2025

Thank you!



Stakeholder comments also welcomed through:

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